PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: GigHub

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysia's freelance economy is severely bottlenecked by a systemic "trust and infrastructure deficit" that paralyses both independent talent and hiring businesses. On one hand, Malaysia's digital freelancers face chronic income instability and are structurally excluded from the traditional banking system, unable to provide a formal financial transcript to secure basic automotive or housing loans. On the flip side, the country's 1.07 million SMEs urgently need this project-based talent to scale affordably, but they are wary of the operational risks posed by poorly defined project scopes, talent ghosting, intellectual property (IP) exposure, and messy disputes without an HR safety net. Ultimately, the core problem is the absence of a neutral technological bridge capable of enforcing clear contract parameters, verifying deliverables, guaranteeing secure payments, and translating informal gig transactions into formalised, bank-verifiable financial histories for both sides of the market.

- **Target Domain:** Malaysian freelancers needing income proof for bank loans, SMEs, and start-up companies without a human resources budget.

- **Proposed Solution Summary:** GigHub operates on an AI-enforced escrow loop that links clear project requirements to guaranteed payments and formal income generation. Before any work begins, the GLM evaluates the client's project brief for clarity, triggering the escrow deposit only once a validated brief is accepted by a freelancer. Upon submission, deliverables are protected via end-to-end encryption while GLM structurally verifies the file's metadata against the original brief, which ensures the client gets exactly what was requested without exposing confidential IP. This automated verification neutralises scope creep, powers data-driven dispute arbitration, and triggers secure fund releases. By locking every project into this verifiable loop, GigHub translates informal gig work into automated, bank-ready "Statements of Earnings", finally bridging independent talent into Malaysia's formal financial system.

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2. Background & Business Objective

- **Background of the Problem:** The Malaysian labour market is undergoing a structural shift where over 3 million individuals now operate as own-account workers, with a rapidly growing segment participating in the professional freelance economy, a trend driven by a 10.1% youth unemployment rate. Despite their economic contribution, nearly 72% of these workers face institutional financial exclusion, as traditional banking algorithms rely on formal "Form EA" or EPF records that freelancers cannot provide, resulting in the systemic rejection of housing and automotive loan applications. Furthermore, while Malaysia's 1.07 million SMEs desperately need this freelance talent to grow, they remain hesitant to engage independent talent due to a lack of automated infrastructure to mitigate operational risks like "talent ghosting" and intellectual property exposure.
- **Importance of Solving This Issue:** It transforms the gig economy from a survival mechanism into a sustainable professional ecosystem, directly advancing Malaysia's goal of high-value digital labour participation. Automating payment security and project validation, it eliminates the legal and financial vulnerabilities that keep the gig workers in a cycle of instability. For the SMEs, solving this issue provides a risk-free path to scale their operations affordably, effectively providing them with a digital HR department that guarantees quality and protects intellectual property. Ultimately, bridging this gap ensures that independent talent can contribute to the nation's GDP while finally gaining access to the fundamental financial rights.
- **Strategic Fit / Impact:** GigHub aligns with the Zhipu AI ecosystem by demonstrating a high-stakes application where the LLM functions as a financial state-machine orchestrator rather than a simple chatbot. GLM-4 is embedded as the non-removable decision layer: if GLM is disabled, the escrow release trigger breaks, brief validation fails, and disputes cannot be arbitrated. The platform directly addresses the domain of AI for Economic Empowerment and Decision Intelligence by delivering quantifiable impact: elimination of non-payment risk, reduction of dispute resolution time from weeks to 48 hours, and generation of formal income documentation that unlocks bank financing for Malaysia's 1.64 million gig workers

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3. Product Purpose

3.1. Main Goal of the System: To provide an AI-enforced escrow marketplace that bridges the trust and infrastructure deficit in Malaysia's gig economy. It aims to empower 1.64 million gig workers and 1.07 million SMEs by automating payment security, formalising income history, and reducing operational risks.

3.2. Intended Users (Target Audience):

- **Freelancers:** Seeking wage protection and formal income proof. Requires bank-ready, standardised PDF income statements and sector-specific benchmarks that prove creditworthiness for mortgages, car loans, and personal financing

- **SMEs/Founders:** Requiring high-fidelity project delivery without administrative friction.

- **Start-up Companies(without HR budget)**: Requires an Autonomous Project Governance Engine that eliminates administrative friction by automating the entire lifecycle of contract compliance, milestone tracking, and escrow-backed payments.

4. System Functionalities

4.1. Description: The system acts as a high-speed generative and verification engine for freelance contracts. It manages the "Escrow State Machine," transitioning projects from funded to released based on objective AI scores.

4.2. Key Functionalities:

- **Structured Brief Builder + GLM Validation**: To prevent unenforceable contracts, companies must complete a structured brief (deliverable type, format, constraints) which **GLM-4** scores for specificity. A brief must score **≥70** before the job is published and escrow is funded.

- **Freelancer acceptance + escrow trigger**: Once GLM approves a brief, it is listed on the marketplace. When a freelancer accepts the job, the platform prompts the company to fund escrow before work begins. This sequence — accept first, then

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escrow — ensures no freelancer starts work without guaranteed payment, while no company is locked into escrow before a match exists.

- **Milestone system**: Project amount broken into up to 5 milestones. 50/50 split option available to ease SME friction. 72-hour auto-release if client doesn't respond to a submission — prevents ghosting. Watermarked previews for design work prevent IP theft before payment.

- **Smart AI Matchmaking**: Two-way matchmaker to connect the right talent with the right project. For freelancers, the AI analyses their resumes, listed experience, and the quality of their past submissions on GigHub to curate a personalised feed of jobs that perfectly fit their specific skills. At the same time, when a company posts a project, the AI instantly scans the platform's talent pool to recommend a shortlist of the most qualified freelancers.

- **Income Intelligence Dashboard**: This feature directly addresses the "unbankable" status of gig workers under the **Gig Workers Act 2025**. It provides monthly earnings summaries, market rate benchmarks by sector, and generates **formal PDF income statements** for bank loan and mortgage applications.

- **Confidential Work (End-to-end encrypted delivery)**: All deliverables on GigHub are treated as confidential by default. Files are encrypted on the freelancer's device before upload (AES-256-GCM), pass through GigHub's servers as an opaque encrypted blob, and are automatically hard-deleted from Cloudflare R2 after 72 hours. GigHub never holds decryptable content.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: GigHub utilizes GLM-4 for all reasoning, scoring, and arbitration tasks. GLM-4 is selected for its high performance in text reasoning at a low cost, making it ideal for processing structured JSON briefs and dispute logs. For visual deliverable verification, the system relies on metadata and activity logs rather than GLM's vision capabilities (glm-4v) to ensure accuracy in design-heavy sectors.

4.3.2. Prompting Strategy:

Prompt Type	System Role	User Turn Input	Expected JSON Output
brief_validation	Structured contract evaluator. Return only valid JSON.	Brief fields as JSON object	{score, gaps[], clarifying_questions[] }
job_matching	Talent matching engine. Return only valid JSON.	Freelancer profile + skills + past job signals + open brief	{ match_score, strengths[], fit_gaps[], recommendation }
income_intelligence	Financial analyst for Malaysian gig economy. Return only valid JSON.	Completed jobs + skills + location + sector	{ income_summary, market_rate_benchmark, income_consistency_score, trend_narrative, recommended_rate_range, income_statement_text }

4.3.3. Context & Input Handling:

GLM-4 has no memory between API calls. All relevant context is assembled server-side and included in each request. Context assembly rules per active prompt type:

- **Brief validation:** full brief JSON including all structured fields, word limits, format requirements, and reference example descriptions
- **Job matching:** freelancer profile object (skills, bio summary, past job signals: completion rate, client approval rate, sectors) + open brief JSON. Runs as a background batch job on brief publish, not per page load

- **Income intelligence:** array of completed job records (title, sector, amount, completion date) + freelancer skill tags + location (city/state) + sector — sourced from jobs and milestones tables; no personal identifiers

Response caching: GLM-4 brief validation scores are cached in Redis with a 24-hour TTL to avoid re-scoring identical briefs. Job matching and income intelligence results are not cached — each call produces a record for the audit trail.

Token budget: all prompts are designed to stay within 2,000 input tokens. max_tokens is set to 1,000 for all calls.

4.3.4. Fallback & Failure Behavior:

Failure Scenario	Immediate Action	Fallback Outcome
GLM returns non-JSON or malformed response	Retry once after 2-second delay	On second failure, flag for human moderator review; milestone held in escrow
GLM API timeout (> 10 seconds)	Retry with exponential backoff (2s, 4s, 8s)	After 3 failures, auto-escalate to human review queue
GLM brief score is borderline (68–72)	Flag for optional human review before publish	Company is shown score and specific gaps; can revise and resubmit
GLM income API fails during PDF generation	Retry once; log error	User notified; PDF generation queued for next available window
Zhipu AI API key rate-limited	Queue GLM calls in Redis job queue	Escrow state unchanged; user sees 'processing' status; no data lost

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5. User Stories & Use Cases

- **User Stories:**
  Freelancer Stories:
  1. As a freelancer, I want to receive guaranteed payment before I start work, so that I am not at risk of non-payment after delivering my output.
  2. As a freelancer, I want the platform to automatically release my escrow if the client does not respond within 72 hours of my submission, so that I am not held hostage by client inaction.
  3. As a freelancer, I want my deliverables to be end-to-end encrypted so that my client's confidential IP is protected and I am not liable for data breaches.
  4. As a freelancer, I want the platform to surface jobs most relevant to my skills and track record, so that I can find suitable work efficiently.
  5. As a freelancer, I want to generate a formal PDF income statement from my completed projects, so that I can submit it to a bank for a housing or car loan application.
  6. As a freelancer, I want to see my earnings benchmarked against market rates in my sector, so that I know whether I am pricing my work competitively.

  Company / SME Stories:
  1. As a company, I want GLM to validate my project brief before it is published, so that I receive deliverables that match my actual requirements and avoid scope disputes.
  2. As a company, I want to fund escrow only after a suitable freelancer has accepted my brief, so that my capital is not locked up before a match is found.
  3. As a company, I want GLM to recommend a shortlist of qualified freelancers when I publish a brief, so that I can fill positions quickly without manual screening.
  4. As a company, I want to review a watermarked preview of creative deliverables before approving escrow release, so that I can evaluate the work without the freelancer's IP being exposed.
  5. As a company, I want a structured dispute process reviewed by a human moderator, so that I can raise legitimate quality concerns with a clear evidence chain.

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6. As a start-up founder without an HR team, I want the platform to manage the entire contract lifecycle autonomously, so that I can hire project-based talent without administrative overhead.

Admin / Moderator Stories:

1. As a platform moderator, I want to review all disputes with the brief, rejection reason, and SHA-256 hash proof-of-delivery as the evidence chain, so that I can issue fair rulings without needing access to encrypted file content.
2. As a platform admin, I want an immutable audit trail of all escrow events, GLM scores, and state transitions, so that I can produce evidence for legal or regulatory inquiries.

- **Use Cases (Main Interactions):**

  Prompt to UI Flow:

User Action	GLM Prompt Fired	UI State During Call	UI State After Response
Company submits brief form	brief_validation	Spinner: 'Validating brief with AI...'	Score badge shown; gap list displayed if < 70; Publish button enabled if >= 70
Brief published (background)	job_matching	Jobs load normally	Jobs ranked by GLM match score; shortlist surfaced to company
Freelancer views marketplace	job_matching (personalised feed)	Jobs load normally	Jobs ranked by GLM fit score for the freelancer
Freelancer clicks Generate Income Statement	income_intelligence	Spinner: 'Building your statement...'	Dashboard updates with benchmarks and consistency score; PDF download button appears

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Core Escrow Flow:

Company posts brief → GLM validates brief (score >= 70) → GLM matches talent (background) → Freelancer accepts → Escrow funded Work submitted (E2E encrypted) → Client decrypts locally → Client reviews → Escrow released └─ Client rejects → Structured rejection form → Human admin ruling → Escrow resolved └─ Client unresponsive for 72 hours → Auto-release fires

Flow Generation:

The flow generation diagram below illustrates the complete GigHub escrow state machine from brief submission through payment release. It covers all three execution paths: the happy path (client approval), the 72-hour auto-release path (client unresponsive), and the dispute path (client rejection). Swimlanes show which actor is responsible for each step.

Key design principles illustrated in the flow:

- GLM-4 gates the entry — no job is published without a brief score of 70 or above
- Escrow is only triggered after freelancer acceptance — no capital lock-up before a match
- Files travel as encrypted blobs only — GigHub never decrypts; the SHA-256 hash is the proof
- Two independent auto-release paths: client approval OR 72-hour clock expiry
- Dispute resolution is human-moderated — GLM-4 has no role in this path
- income_intelligence fires asynchronously after escrow release to update the earnings dashboard

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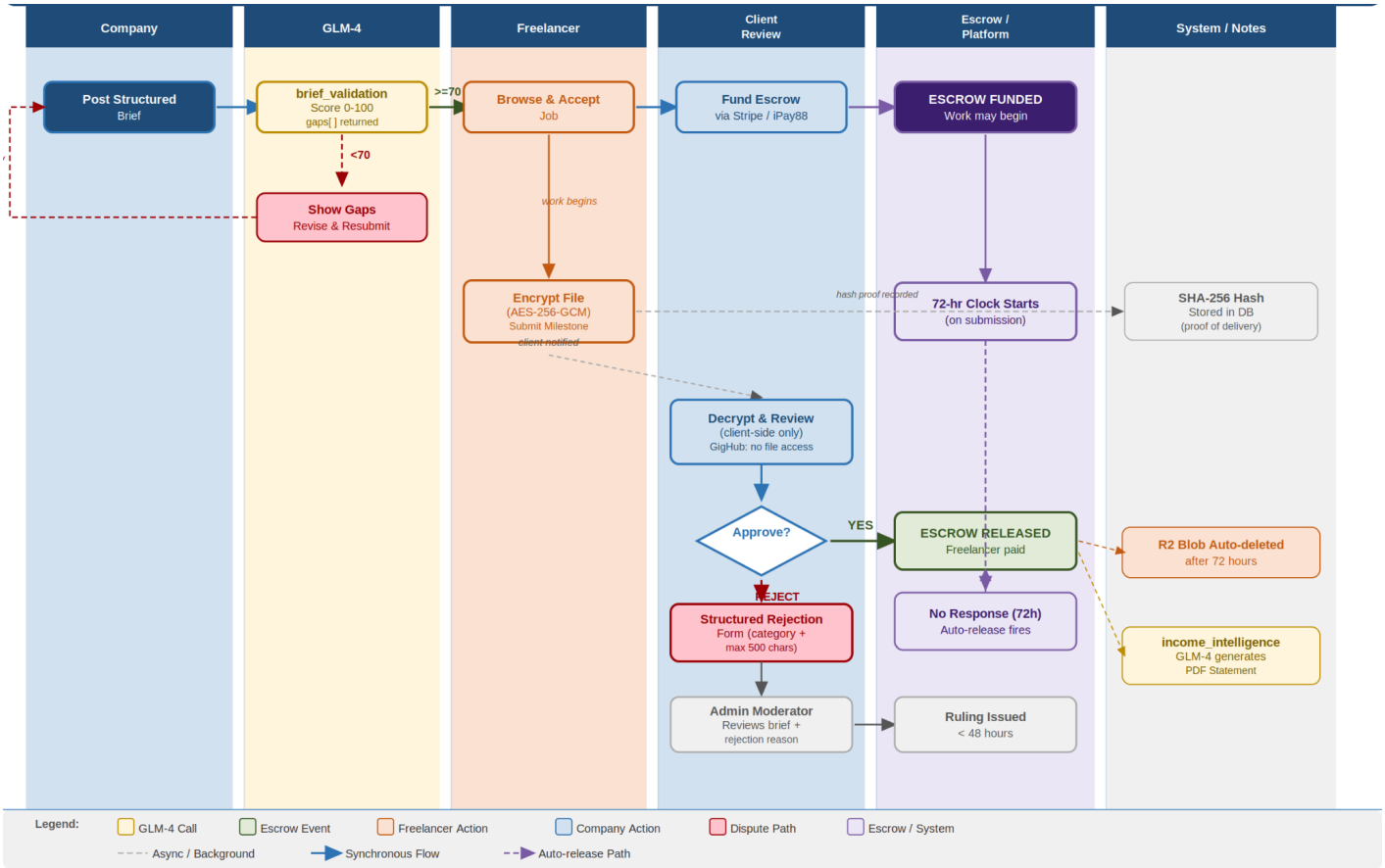

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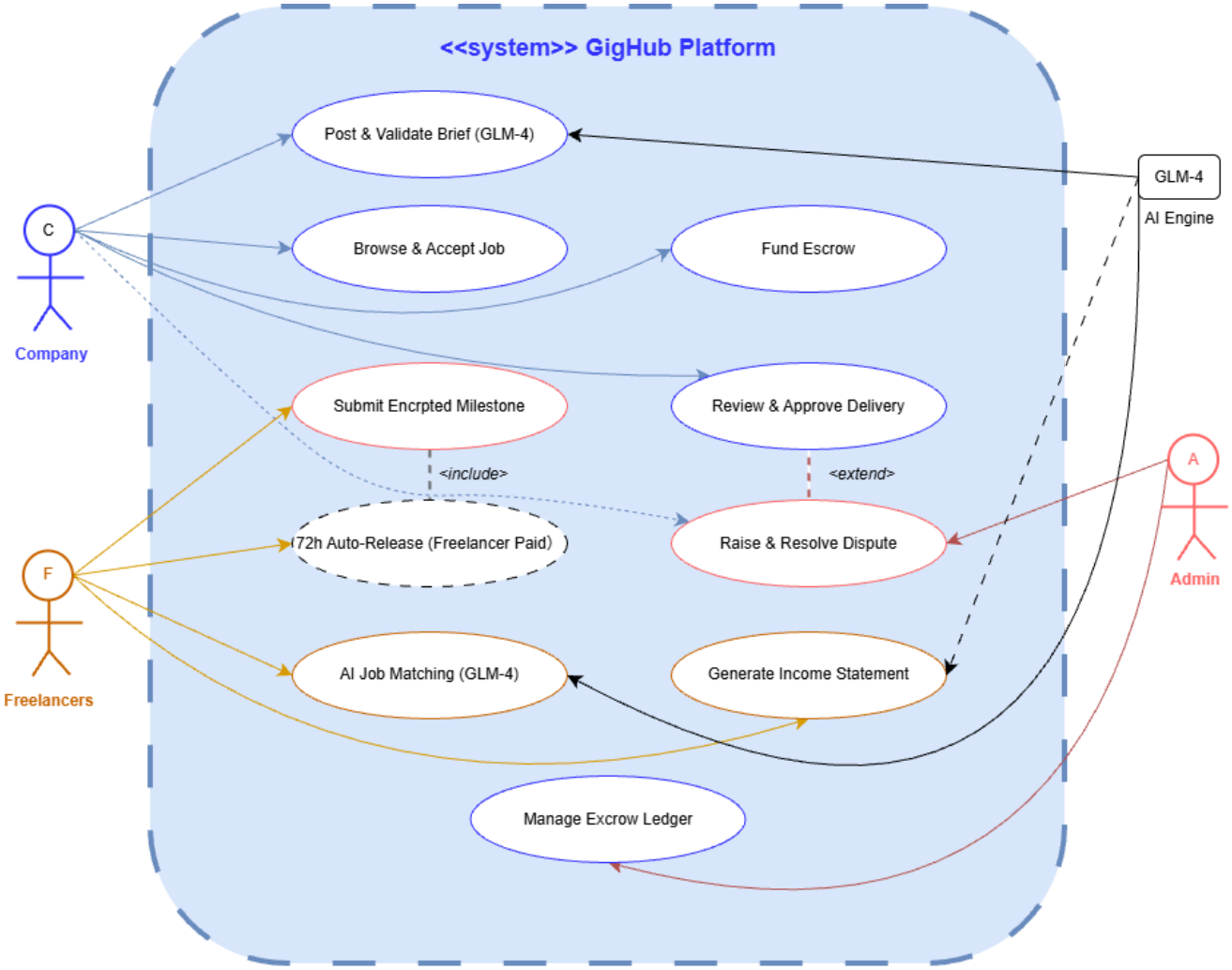
Use Case Diagram:



6. Features Included (Scope Definition)

Feature	Priority	GLM-4?
Structured brief builder with form validation	Must have	Partial
GLM-4 brief validation and score gate (≥70)	Must have	Yes — core gate
Freelancer marketplace (browse and accept jobs)	Must have	No
GLM-4 job matching — personalised feed + company shortlist	Must have	Yes
Freelancer acceptance → escrow funding trigger	Must have	No
Escrow ledger (append-only PostgreSQL)	Must have	No
Milestone creation and management (up to 5)	Must have	No
Client-side AES-256-GCM file encryption (Web Crypto API)	Must have	No
Encrypted file upload to Cloudflare R2	Must have	No
SHA-256 hash proof-of-delivery recording	Must have	No
72-hour auto-release mechanism	Must have	No
Watermarked preview for design deliverables	Should have	No
Client-side decryption and review flow	Must have	No
Structured client rejection form (categorised reasons)	Must have	No
Admin dispute moderator queue (48-hour SLA)	Must have	No
Income Intelligence dashboard (earnings summary)	Must have	Yes (GLM benchmarking)
GLM-4 market rate + consistency score + trend narrative	Must have	Yes

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PDF income statement generation (pdfkit)	Must have	Yes (GLM generates text
72-hour R2 lifecycle auto-delete (encrypted blobs)	Must have	No
Freelancer profile + portfolio + skill taxonomy	Must have	No
JWT authentication + role-based access control	Must have	No
Stripe sandbox payment gateway	Must have	No
Real-time notifications (Redis queue)	Should have	No

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7. Features Not Included (Scope Control)

Excluded Feature	Reason
GLM-4 milestone content verification	Removed: E2E encryption means GigHub never holds plaintext so GLM-4 cannot assess file content
GLM-4 dispute arbitration	Removed: without content visibility GLM-4 has no basis for a ruling; disputes are human-moderated
Mobile native app (iOS / Android)	Web app is mobile-responsive via Next.js; native app adds complexity for no hackathon demo value
FPX live payment	Stripe sandbox covers payment for demo; live Malaysian gateway requires MSB licensing
Multi-currency support	The platform operates in MYR only for MVP
MyKad identity verification	IC stored encrypted as a profile field; live e-KYC requires BNM licensing
EPF i-Saraan integration	Income statements reference EPF gap as advisory; EPF API not available to third parties
In-platform messaging / chat	All communication scoped to structured forms to maintain audit trail integrity
Bahasa Malaysia UI localisation	MVP ships in English; BM localisation is post-hackathon
QR code statement verification endpoint	Requires live public URL; post-hackathon
3-month income projection	Needs sufficient historical data; skip for prototype
Subscription billing for companies	Revenue model designed but not implemented in MVP

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8. Assumptions & Constraints

- **GLM Cost Constraint:**
  - Every AI prompt costs money per token. Processing long chat histories for disputes could become expensive.
- **Technical Constraints:**
  - Frontend: Next.js (App Router) on Vercel free tier. Web Crypto API required — Chrome 60+, Firefox 57+, Safari 11+.
  - Backend: Node.js + Express. All GLM-4 prompts server-side only. API key never exposed to browser.
  - Database: PostgreSQL (Supabase free tier). Row-level security via user_id. Escrow transitions use PostgreSQL transactions.
  - File storage: Cloudflare R2. Ciphertext only. R2 lifecycle rule + cron for 72-hour deletion.
  - Payments: Stripe test mode for hackathon. Live processing requires Malaysian MSB licence.
  - Escrow ledger is append-only — no UPDATE on escrow_ledger rows.

- **Performance Constraints:**
  - GLM-4 latency: 2–6 seconds. Brief validation synchronous with UI loading state. Matchmaking asynchronous.
  - 72-hour auto-release via cron every 15 minutes — not real-time.
  - PDF generation 5–15 seconds, handled asynchronously with frontend status poll.
  - R2 file upload limit: 100 MB per file for MVP.

- **User Input:**

  - Brief form: all fields required. Max 2,000 characters per field. References are text descriptions only.
  - Rejection form: category from structured list + max 500 characters elaboration.
  - Milestone amounts must sum exactly to job.total_amount.
  - Skill tags from predefined taxonomy to ensure GLM-4 matching quality.

9. Risks & Questions Throughout Development

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Design Similarity:

1. **Platform similarity risk**
   - Risk: GigHub's escrow model resembles Upwork at the surface level.
   - Mitigation: Differentiation is the GLM-4 decision layer and E2E encryption architecture. On Upwork, escrow release is client-discretionary; on GigHub, it is time-gated by 72-hour auto-release. The income intelligence PDF for the Malaysian market has no equivalent in global platforms. Demo should make GLM non-removability explicit to judges.
2. **Dispute resolution without GLM**
   - Risk: Without GLM-4 in the dispute path the platform relies on human moderation.
   - Mitigation: Frame as a strength. E2E encryption is a genuine PDPA and IP protection feature. Because GigHub never holds plaintext, human moderation is the only legally defensible dispute process. The 72-hour auto-release prevents most disputes from arising.
3. **Brief score threshold calibration**
   - Risk: The ≥70 threshold is a design assumption.
   - Mitigation: Run calibration batch of 20 test briefs during development. Threshold stored as a configuration constant, not hardcoded.

Frame Stability:

1. **Avoid AI auto-approval for media**
   - Risk: During development, we considered using AI to automatically approve video or animation files by reading their metadata. However, the risk is that metadata cannot verify "frame stability"—meaning an AI cannot detect if an encrypted video is visually glitchy or contains blank black frames
   - Mitigation: To eliminate this risk, GigHub intentionally delegates milestone approval to a manual client review. The SME has a 72-hour window to locally decrypt the file, watch the video, and manually approve or revert the submission if it does not meet the requirements.

Drawing Interpretation:

1. **Avoid AI auto-approval for technical work**
   - Risk: For technical drafting (like AutoCAD blueprints), a major risk is that an AI cannot interpret mathematical dimensions or structural layers once a file is end-to-end encrypted. Relying on AI here could result in unsafe or mathematically incorrect work being automatically approved and paid for.

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- Mitigation: GigHub's architecture mitigates this by keeping the AI out of the file-checking process. The platform relies entirely on the SME to manually inspect the drawing's math and measurements. If it is wrong, the SME simply clicks "Reject" to force the worker to resubmit, keeping full quality control in human hands.

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